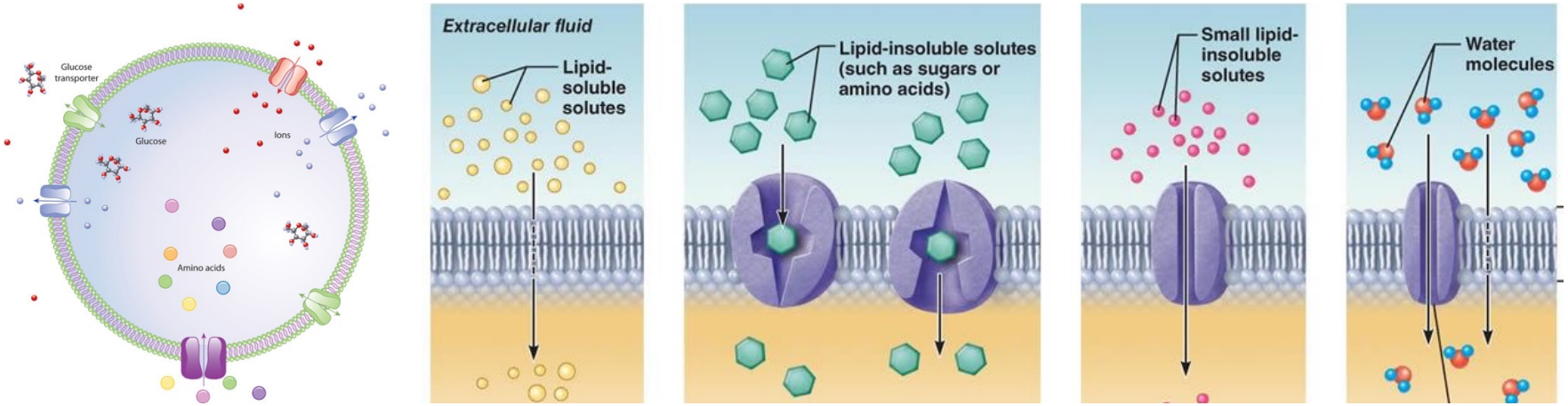
Microtransport through the membrane

Lecture 3.3 September 9th, 2026



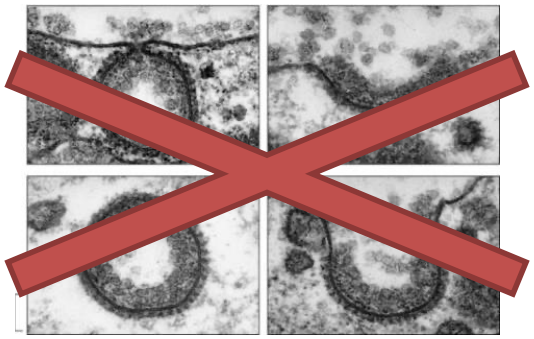
Dr. Serrano | dserrano@unav.es

1 Introduction

Transport through the plasma membrane

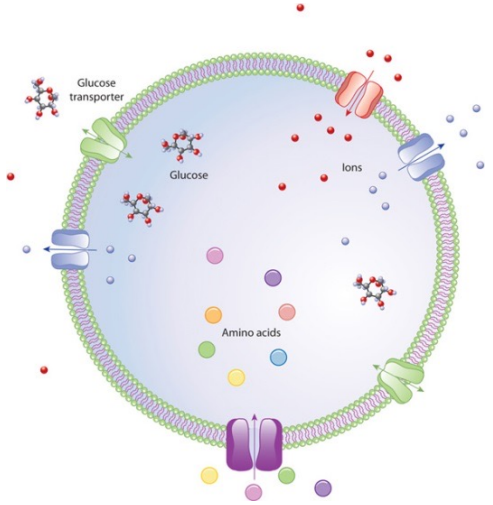
Microtransport

- No deformation of the plasma membrane



Macrotransport => Lecture 3.4

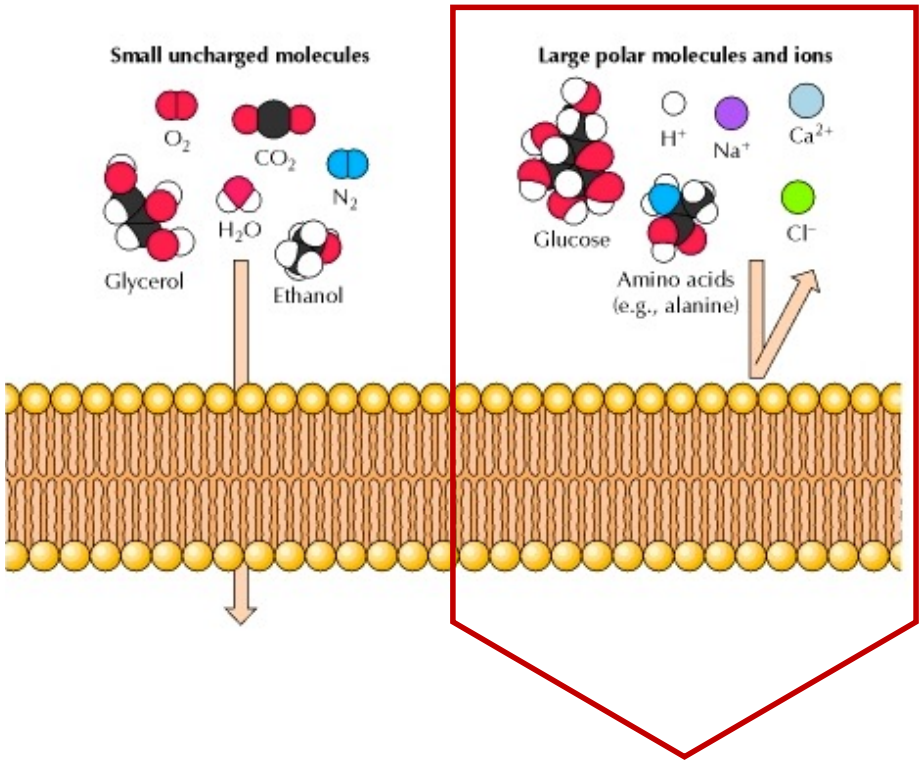
- Examples: transport of ions, H2O and small molecules
- Types:
  - PASSIVE TRANSPORT (no energy)
    - a) Simple diffusion
    - b) Facilitated diffusion
  - ACTIVE TRANSPORT (requires energy)
    - a) Direct
    - b) Indirect



2 Passive transport

- **No energy required.**
- **Down-gradient**: from high concentration to low concentration.
- Through plasma membrane or facilitated by transporters.

2.1 SIMPLE DIFFUSION



- FACILITATED DIFFUSION
- ACTIVE TRANSPORT

Affected by:

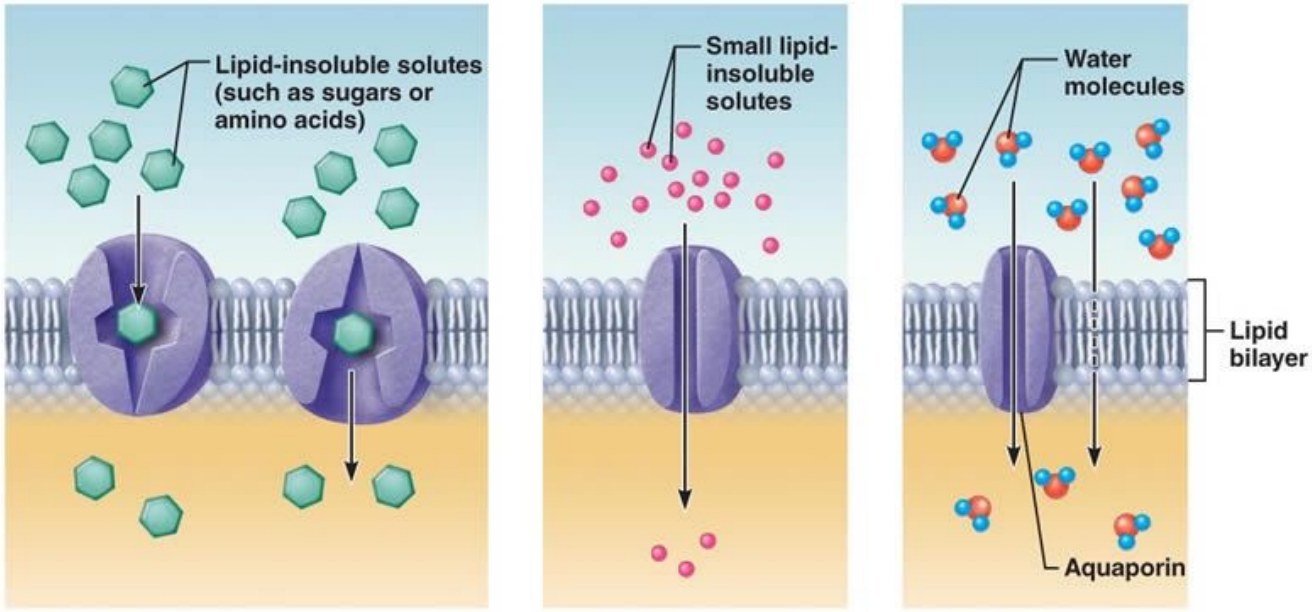
- **Size** ↓
- **Polarity** ↓
- **Liposolubility** ↑
- **Gradient (concentration)** ↑
- **Charge** ↓

- **Gases:** O_2, CO_2, N_2
- **Hydrophobic substances:** benzene
- **<u>Small</u> hydrophilic uncharged polar:** water, ethanol, glycerol…

2 Passive transport

- **No energy required.**
- **Down-gradient**: from high concentration to low concentration.
- Through plasma membrane or facilitated by transporters.

2.2 FACILITATED DIFFUSION



(b) Carrier-mediated facilitated diffusion via protein carrier specific for one chemical; binding of substrate causes transport protein to change shape

(c) Channel-mediated facilitated diffusion through a channel protein; mostly ions selected on basis of size and charge

(d) Osmosis, diffusion of a solvent such as water through a specific channel protein (aquaporin) or through the lipid bilayer

Facilitated **by proteins**

1. **Carriers (permeases):** glucose.
2. **Membrane channels:** ion channels, aquaporins.

- **Regulated**
- **Selective** (Na+, K+, H+, Glucose, water…)
- Saturation:
  - Carriers: saturation
  - Membrane channels: no saturation

2 Passive transport

- **No energy required.**
- **Down-gradient**: from high concentration to low concentration.
- Through plasma membrane or facilitated by transporters.

2.2 FACILITATED DIFFUSION - REGULATION

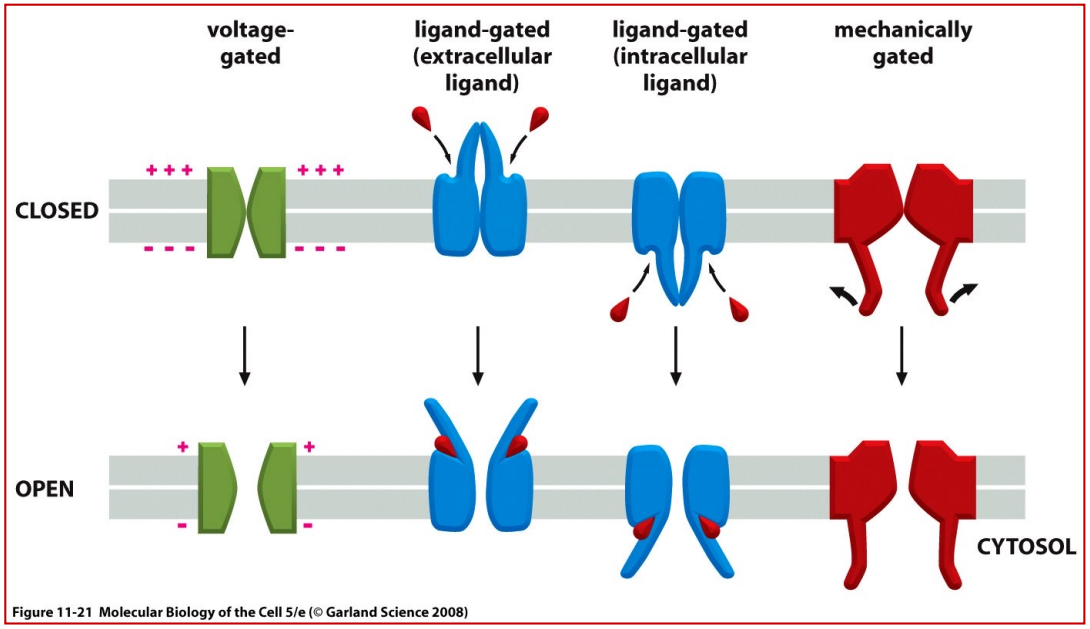
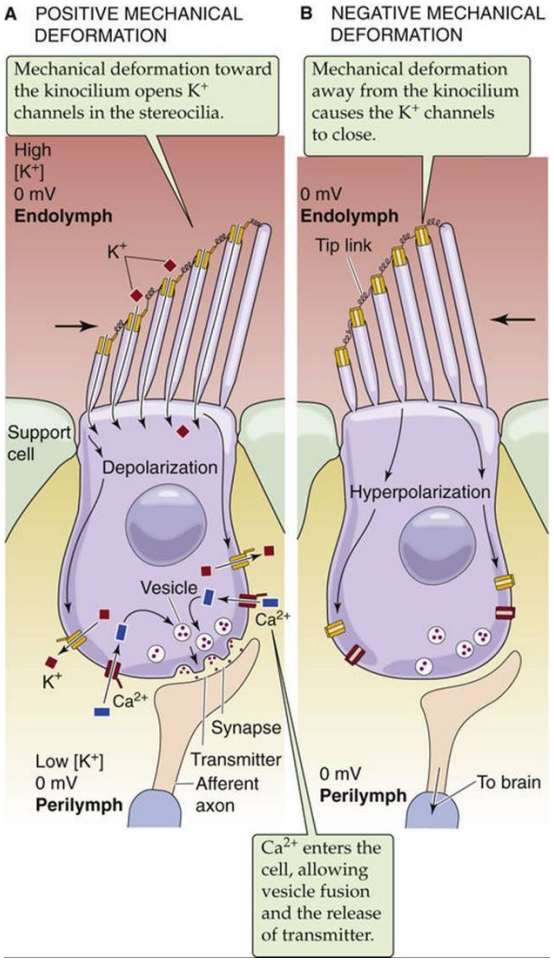


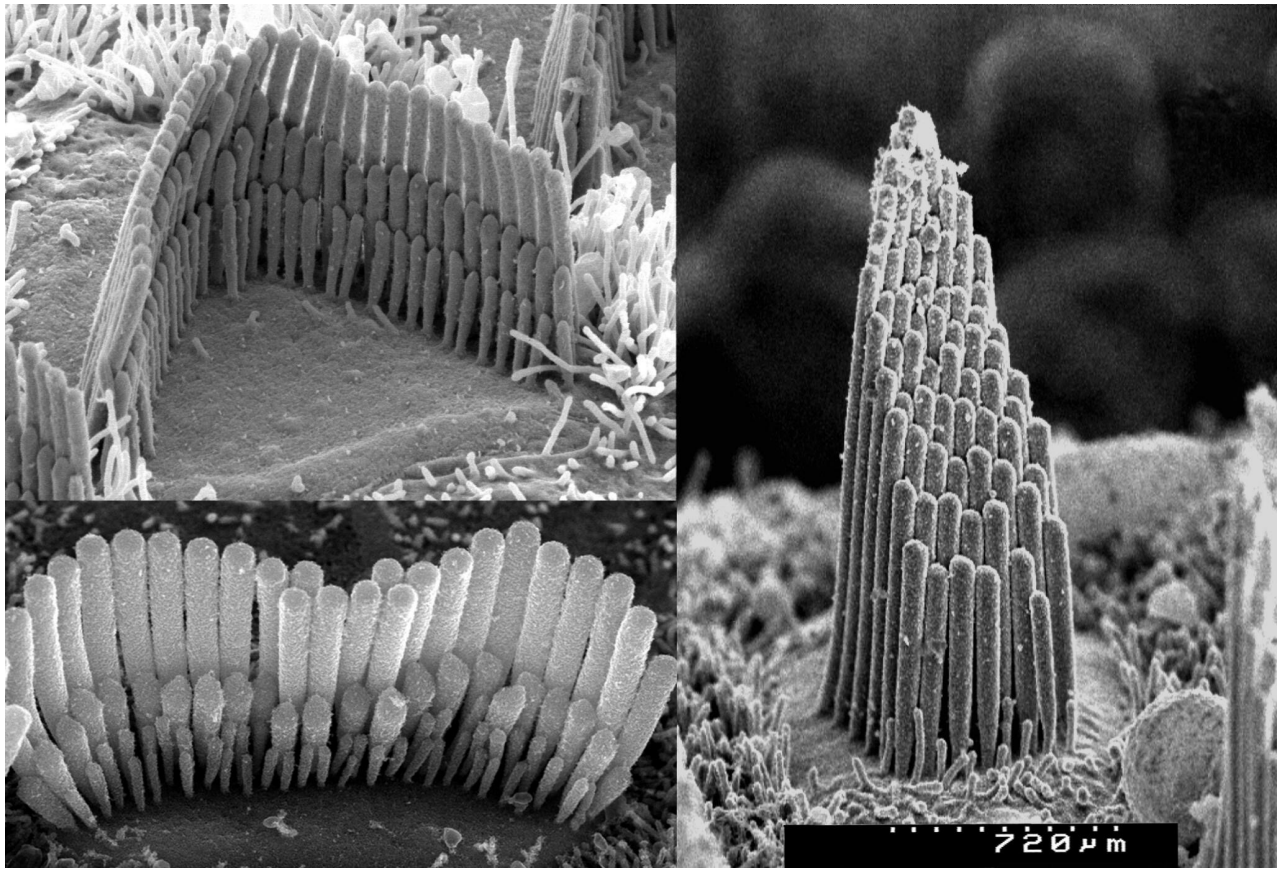
Figure 11-21 Molecular Biology of the Cell 5/e (© Garland Science 2008)

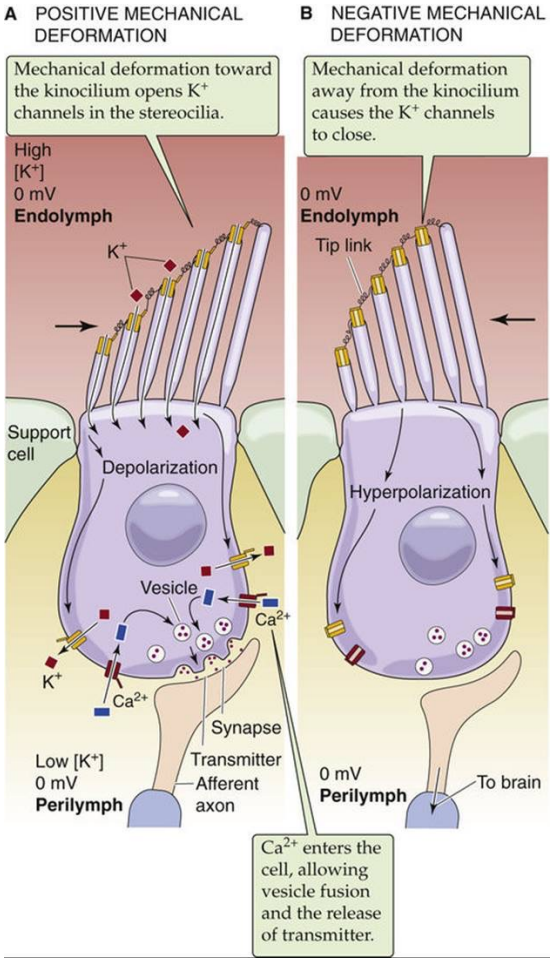


2 Passive transport

- **No energy required.**
- **Down-gradient**: from high concentration to low concentration.
- Through plasma membrane or facilitated by transporters.

2.2 FACILITATED DIFFUSION - REGULATION





2 Passive transport

- **No energy required.**
- **Down-gradient**: from high concentration to low concentration.
- Through plasma membrane or facilitated by transporters.

2.2 FACILITATED DIFFUSION - REGULATION

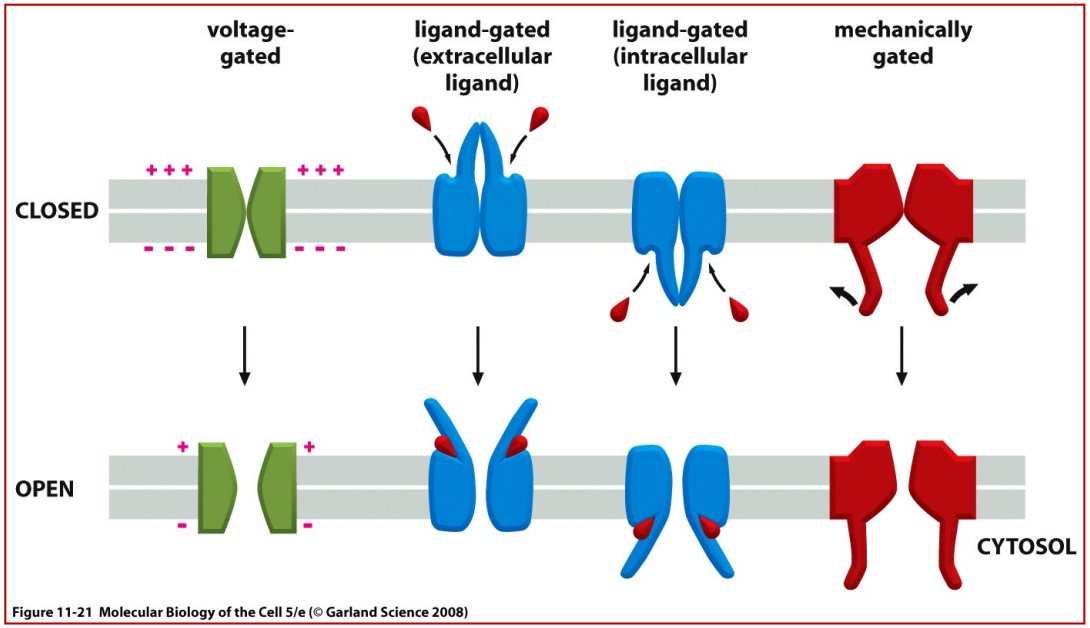
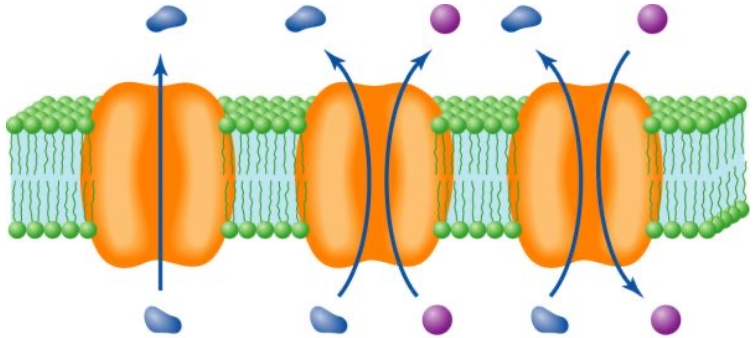


Figure 11-21 Molecular Biology of the Cell 5/e (© Garland Science 2008)



uniport — Glut: glucose transporter

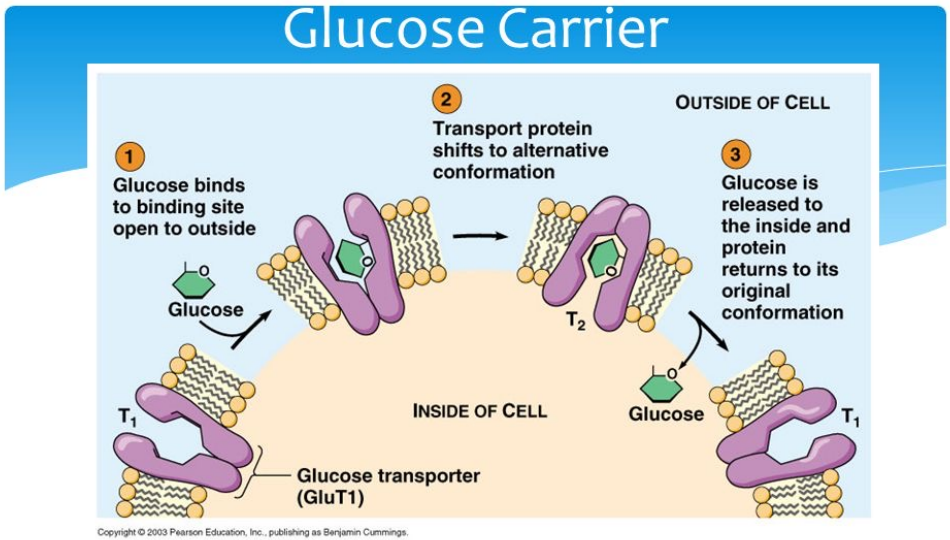
symport — Na+/Glucose

antiport — HCO_3- / Cl-

2.3 Examples of passive transport

Permeases

- Down the gradient
- **Binding to the solute**
- **Conformational change**
- Saturable
- Specific
- Regulable



- Glucose carrier crosses the membrane and has at least 2 conformations.
- One conformation exposes the binding site on the outside of the cell and the other on the inside of the cell.

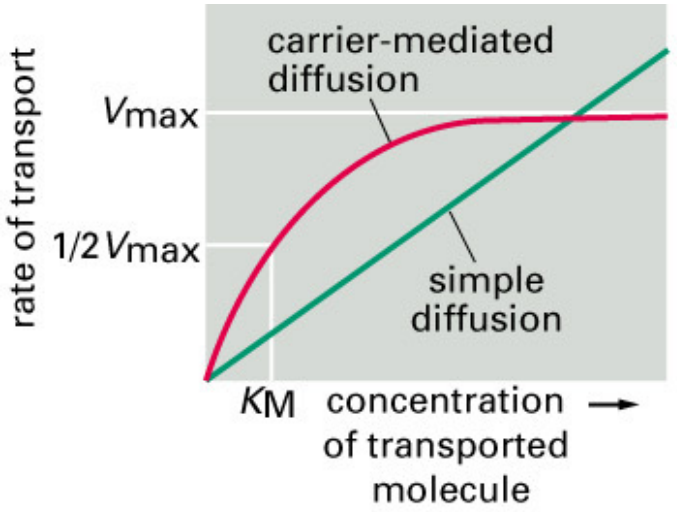
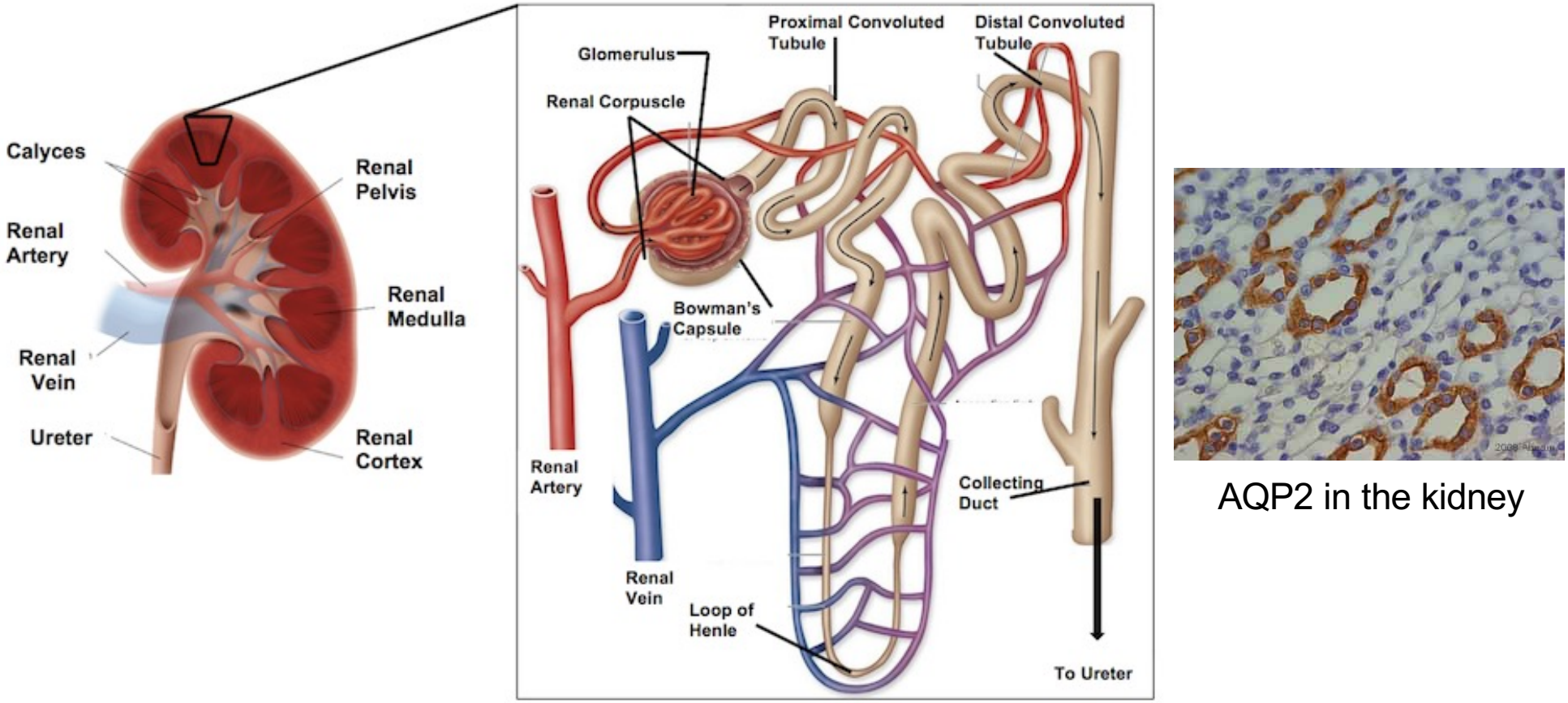


Figure 11–7. Molecular Biology of the Cell, 4th Edition.

2.3 Examples of passive transport

H_2O Transport

Renal filtration: **180L of blood per day**; reabsorption 178L; **2L urine per day**

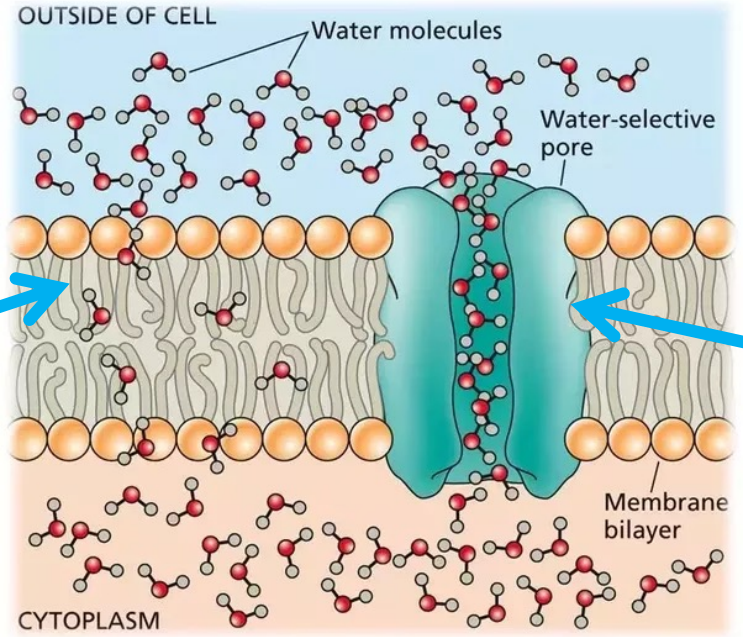


AQP2 in the kidney

2.3 Examples of passive transport

H_2O Transport

Simple diffusion





Peter Agre
Nobel prize in Chemistry 2003

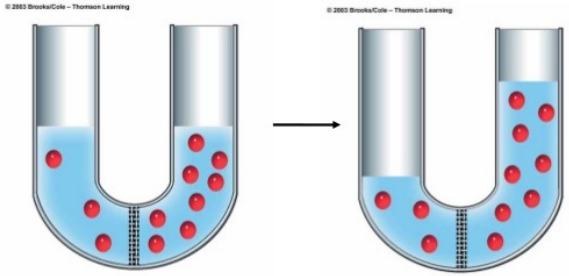
Facilitated diffusion Aquaporins

- Selective channels for H_2O transport
- Abundant in organs like the kidney

Osmosis: Diffusion of H_2O across a semi-permeable membrane

As the concentration of solute increases the concentration of solvent (H_2O) decreases

Compared to an equal volume of water, the solute/H_2O mixture has less H_2O molecules as that space is taken up by the solute

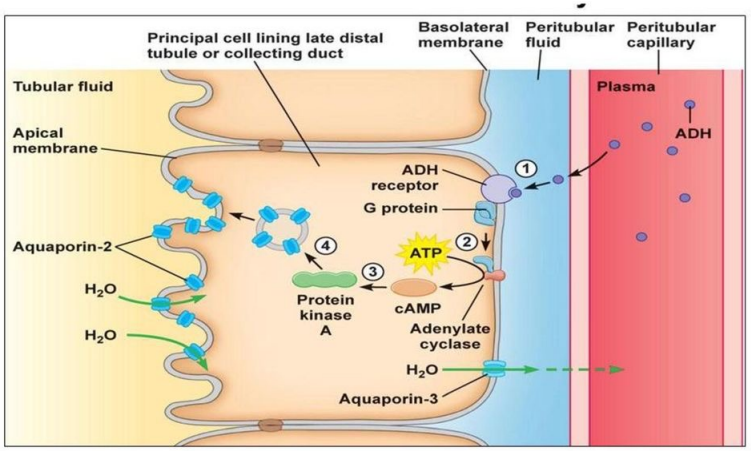


2.3 Examples of passive transport

H_2O Transport – Regulation of AQP2



Aquaporin 2 *(AQP2)*



Mechanism: ↑ **cAMP** in cells & Formation of microtubules in cell membrane **(Aquaporin 2).**

Diabetes insipidus

≠ diabetes mellitus

↓

Alteration of AQP2 or ADH

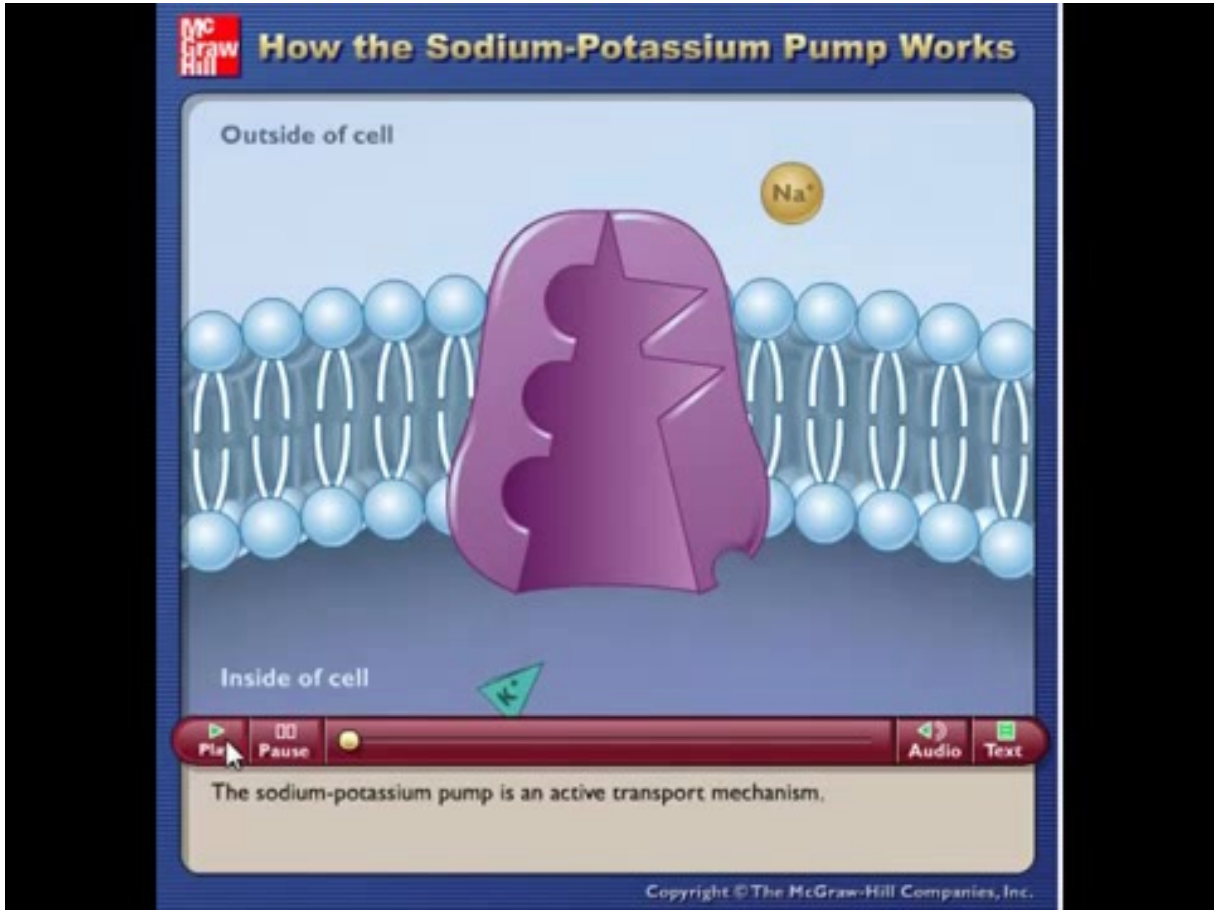
↓

-Polyuria
-Thirst
-Dehydration
-Hypovolemia

❸ Active transport - DIRECT

- **Against gradient**
- **ATP** expenditure
- It **binds** to the solute
- **Conformational change**
- Get **saturated**
- **Selective**
- **Regulated**

Example:
Na+/K+ pump



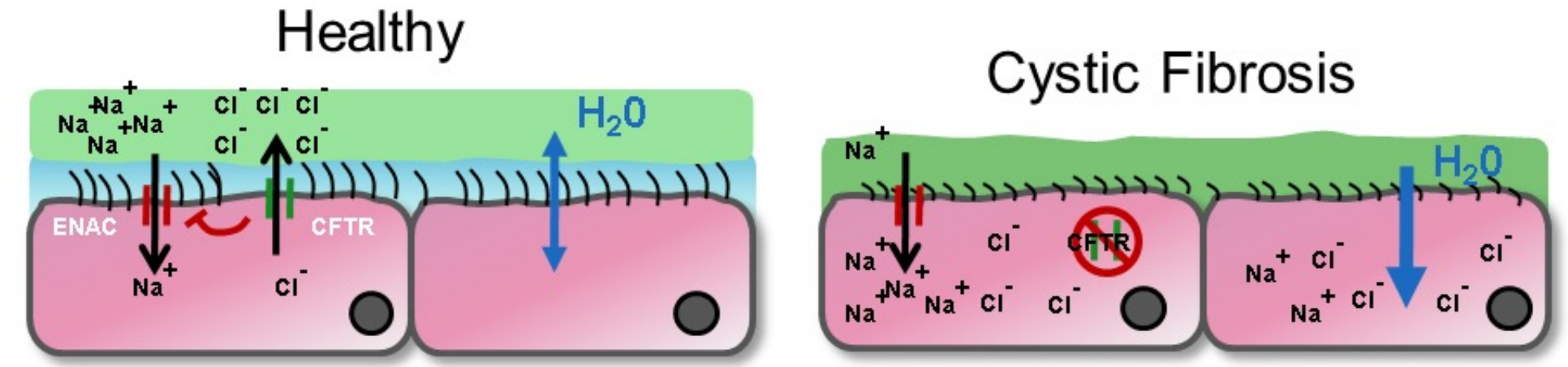
3.1 ABC Transporters

» ABC: *ATP binding cassette*

» **Functions:** incorporation of molecules into the cell and release of xenobiotics, **drugs**, toxic agents…

MDR1 (*Multi-drug resistance*)
MDR2
ABCG2
} Resistance to chemotherapy

Example: CFTR protein (*Cystic fibrosis transmembrane conductance regulator*): Active transport of Cl-



3.2 Cystic fibrosis

-No cure

-Life expectancy: <40 years.

-Treatment:

Preventing and controlling infections.

Removing mucus from the lungs.

Lung transplant

Preventing intestinal blockage.

Providing adequate nutrition.

-Gene therapy 

-2019: **breakthrough !!!!**



Francis Collins

Realizing the Dream of Molecularly Targeted Therapies for Cystic Fibrosis

Francis S. Collins, M.D., Ph.D.

The diary entry of an 8-year-old girl with cystic fibrosis indicates that Aug. 25, 1989, was an important day for her (Fig. 1). That was the day the research teams at the University of Michigan and the Hospital for Sick Children, Toronto, announced the discovery of the cystic fibrosis gene and the most common mutation, a three-base deletion that results in a missing phenylalanine in codon 508 (denoted the Phe508del *CFTR* mutation). We hoped that the gene discovery would someday lead to effective treatments for children and adults with cystic fibrosis, but we knew that

N ENGL J MED 381;19 NEJM.ORG NOVEMBER 7, 2019

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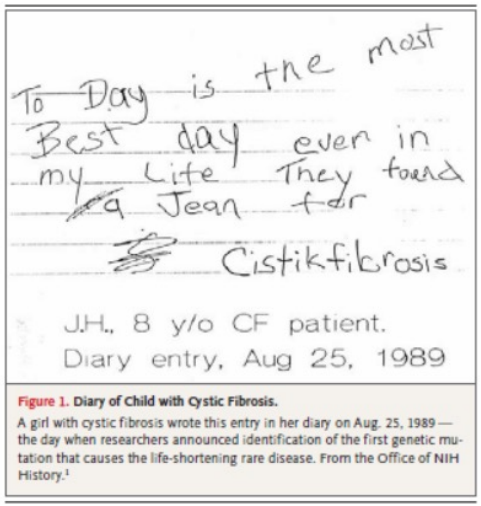


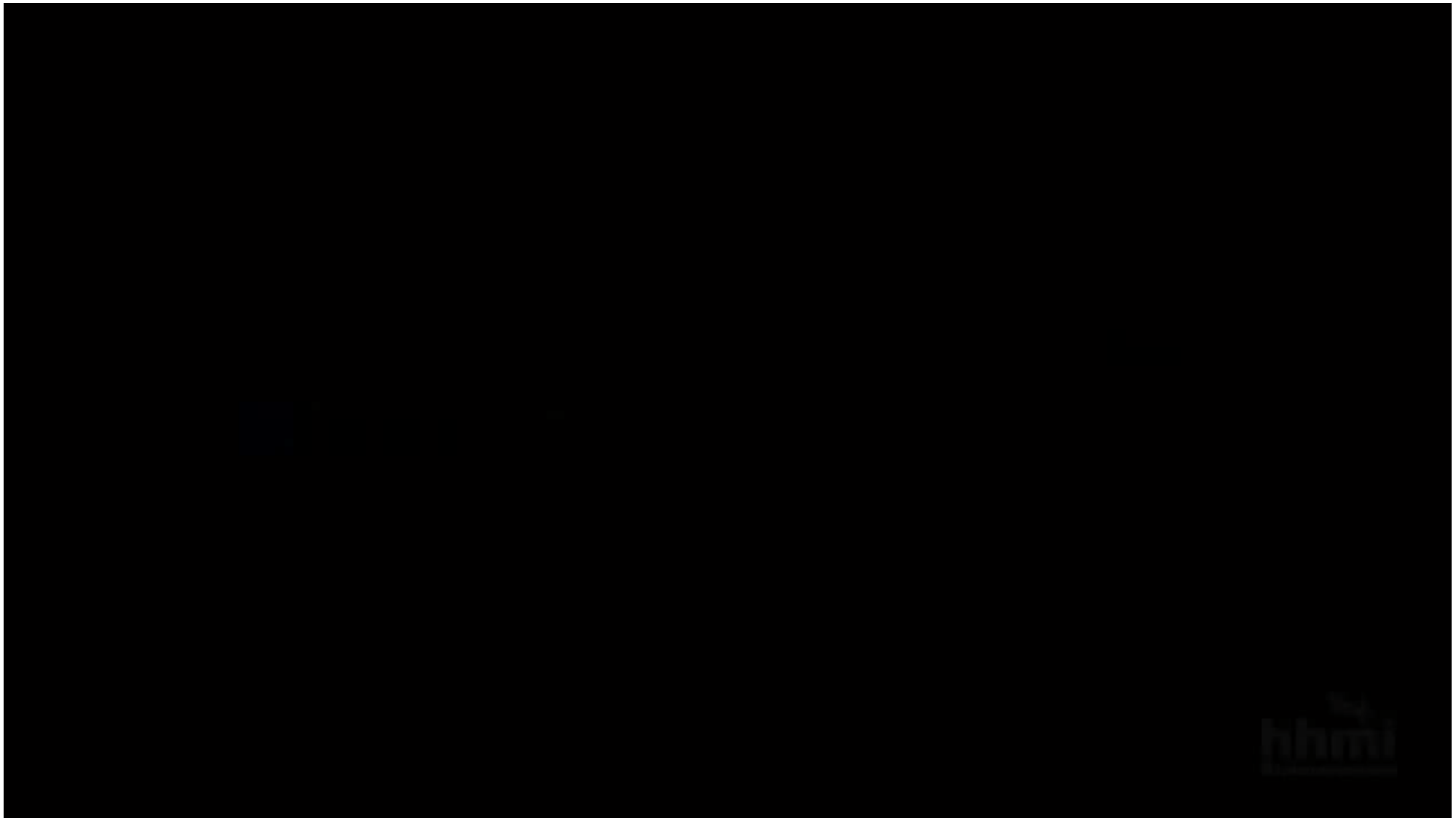
Figure 1. Diary of Child with Cystic Fibrosis.

A girl with cystic fibrosis wrote this entry in her diary on Aug. 25, 1989 — the day when researchers announced identification of the first genetic mutation that causes the life-shortening rare disease. From the Office of NIH History.[1]

testinal tract, sweat glands, and other organ systems. Drug researchers had to consider that the Phe508del *CFTR* mutation, the most common of the more than 1700 known *CFTR* mutations that can cause cystic fibrosis,[7] results in a protein with not just one — but two — serious functional deficits. Not only do patients with this mutation have a CFTR protein that is misfolded and trapped in the endoplasmic reticulum of the cell, but any CFTR that does manage to reach the proper location in the cell membrane is deficient in activation. So the search for therapeutics had to include both small molecules that could correct the misfolding of the CFTR protein ("correctors") and those that could activate its function when it reaches the cell membrane ("potentiators").

Academic investigators, many funded by the National Institutes of Health, derived a deep understanding of CFTR function in the first 10 years after gene discovery. Building on that fundamental platform of knowledge, the Cystic Fibrosis Foundation and a small company called Aurora Biosciences (later, Vertex Pharmaceuticals) joined forces in the late 1990s to embark on a high-

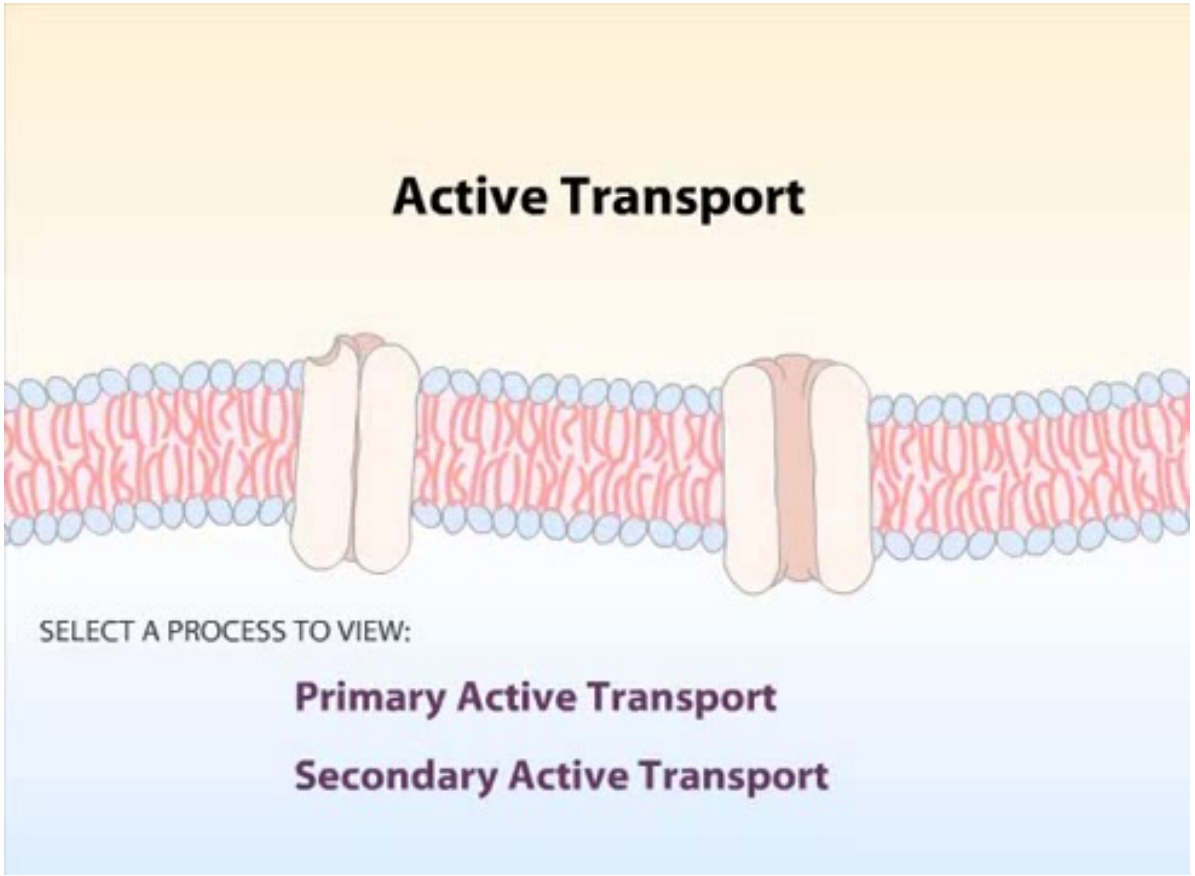
3.2 Cystic fibrosis



4 Active transport - INDIRECT

Characteristics are similar to those of direct active transport, but they **do not use directly ATP** as the energy source

Example: Sodium/glucose linked transport (SGLT)



5 Summary

	PASSIVE		ACTIVE	
	SIMPLE	**FACILITATED**	**DIRECT**	**INDIRECT**
Protein involved	No	Yes (Carriers or Channels)	Yes	Yes
Energy	No	No	YES	YES (indirect)
Selective	No	Yes	Yes	Yes
Regulation	No	Yes	Yes	Yes
Reversible (bidirectional)	Yes	Yes	No	No
Saturation	No	Carriers: Yes Channels: No	Yes	Yes